Validation of the analytical turbulence-averaging formulas

Twin-Field QKD over free-space optical channels: log-normal (Hermite / Euler-Stirling) and Gamma-Gamma (cumulant) expansions

Test of the formulas in the manuscript "*Analytical Performance Analysis of Twin-Field QKD over Turbulent FSO Channels*" and in the progress report "*Turbulence Averaged Performance of Twin Field QKD over Free Space Optical Channels*" (Aug 22 - Sep 30, 2026).

Generated 12 October 2026 by **analytical_validation/** (scripts 01, 02, 03). All numbers in this document are read from **results/tables/**; nothing is typed in by hand.

Script	What it tests	PASS	FAIL	INFO
01_test_lognormal_formulas.py	Log-normal framework: derivatives, Euler operator, X- vs η-space, Tables IV-VI, moments, stress test, Monte Carlo	40	0	7
02_test_gammagamma_formulas.py	Gamma-Gamma cumulant expansion with aperture averaging: cumulants, Tables 1-2, Figs 1-3, expansion point, references	30	0	6

Operating point (manuscript Table III): $\alpha = 0.2$, $p_d = 10^{-7}$, $\theta_A = -\theta_B = \arcsin\sqrt{0.02}$, $N_{max} = 4$, $\varphi = 0$, $\Omega = 0.96$, $C_n^2 = 10^{-17}$ $m^{-2/3}$, $\lambda = 1550$ nm, $w_0 = 5$ cm, receiver radius 15 cm, 0.2 dB/km. Convention: $\eta = \sqrt{(\eta_A \eta_B)}$ with $\gamma = \alpha^2 \eta$; the validated engine takes $\eta_A \eta_B = \eta^2$.

Reference methods (never the formulas under test): Gauss-Hermite and adaptive quadrature in $X = \ln(\eta/\eta_0)$ for log-normal; tensor Gauss-Laguerre quadrature, direct Bessel-K density quadrature and Monte Carlo for Gamma-Gamma; a 60-digit mpmath re-implementation with numerical differentiation for the analytic derivatives.

1. Summary of findings

What holds. Every accuracy claim of the formulas themselves is reproduced: the analytic derivative machinery (Appendix A-H) matches an independent 60-digit reference to ~10^{-10} at all orders; the X-space and η-space routes agree (78 cross-checks); all 54 entries of manuscript Tables IV-VI are reproduced within rounding; the 4-term log-normal key-rate error stays below 0.09 % on the 13-point sweep; and the Gamma-Gamma 4th-order formula stays within ~0.02 % of quadrature in every row, including Rytov variance 5, under either parameter convention. Using the log-normal formula on a Gamma-Gamma channel costs more than 50 % of the key rate at Rytov variance 5, and raising its order does not repair this.

#	Finding	Where	Effect
1	**Gamma-Gamma exponent in the progress report.** Fig. 2 (SI peak 0.71, decaying to ~0.08 at Rytov variance 10) and the Table 1/2 entries for Rytov variance > 1 are reproduced only if $\sigma_R^{12/5}$ is evaluated as (Rytov *variance*)$^{12/5}$. In Andrews-Phillips σ_R is the standard deviation (exponent 6/5 on the variance); then the scintillation index peaks at ~1.2-1.3 and saturates toward 1. Same slip as in the repository code before v2.2.0.	Report Sec. 3.3, Figs 2-3, Tables 1-2	Cumulant formulas unaffected. Aperture gain and delta-method columns change for $\sigma^2 > 1$ (e.g. gain at 160 km: +2.9 % vs +4.7 %). 4th-order error < 0.03 % either way.
2	**p_{XX} bracket.** Eq. (6) and Appendix A print the A^2 term inside the factor 1/2. CAL Eq. (30) and the engine have $p_{XX} = (A/2)[e^{-(1-\Omega)\gamma} + e^{-(1+\Omega)\gamma}] - A^2e^{-2\gamma}$. As printed it gives 0.5 instead of 0.0016 at 80 km.	Manuscript Eq. (6), App. A	Typo only; the corrected form matches the reference to ~10^{-16}.
3	**D_j is η-independent.** App. E says $D_j^{(n)}$ "follows identically" from the $\sqrt{p_{ZZ}}$ terms; in CAL and the engine D_j is a constant, so its derivatives vanish.	Manuscript App. E	Using the constant D_j reproduces the reference to ~10^{-10}.
4	**Typo "$\sigma_X^2 = 9.6$".** The 29 % naive-method error quoted for $C_n^2=10^{-16}$, L=140 km is reproduced (28.8 %) at $\sigma_X^2 = 2.40$; $9.6 = \sigma_A^2+\sigma_B^2 = 4\sigma_X^2$.	Manuscript App. H	Typo only.
5	**L = 20 km row of report Tables 1-2.** The same +0.36 % appears in all four columns (delta, 2nd, 3rd, 4th order), so it cannot come from the formulas. With the Laguerre reference (confirmed by Bessel-K quadrature and Monte Carlo) all four columns are within 0.006 % there.	Report Tables 1-2	Likely a reference-integral issue at very large shape parameters (α, β = 43, 162).
6	**Agreement of the two routes:** the manuscript says "better than 10^{-9}"; the worst of the 78 checks here is a few 10^{-9}, limited by double-precision cancellation in the analytic derivatives of e_X, e_Z.	Manuscript Sec. VI-B	Cosmetic. Say "~10^{-8}" or evaluate in extended precision.
7	**"Median" for Gamma-Gamma.** exp(E[ln η]) is the geometric mean (log-mean), not the median; the two coincide only for log-normal. The conclusion (this point beats the mean and the mode) is confirmed.	Report Sec. 4.3	Terminology.
8	**Aperture term in α.** The report applies aperture averaging to β only; Andrews-Phillips also have a d^2 term in α. Worth stating as a deliberate simplification (verify against ref. [3]).	Report Sec. 3.3	Small effect; not tested separately.

2. Log-normal framework (manuscript)

Formulas under test: $E[Q] = Q(\eta_{med}) + (\sigma_{X_2}^2/2)\,\Theta^2 Q + (\sigma_X^4/8)\,\Theta^4 Q$, with $\Theta = \eta\, d/d\eta$ expanded through Stirling numbers (Eq. 34-40), applied to e_X, e_Z, R with the analytic derivatives of Appendix A-H. $\eta_{med} = \eta_0 e^{-\sigma_X}$, $\sigma_X^2 = \sigma_R^2(L/2)/2$.

2.1 Geometry: validity distance and Rytov variance (Tables I-II, Eq. 15, 21)

Cn^2 [$m^{-2/3}$]	validity L, solved [km]	printed Table I [km]	Eq. (21) [km]
1e-17	208.80	208.8	208.80
1e-15	16.94	16.9	16.94
1e-13	1.37	1.37	1.37

Cn^2	L [km]	$\sigma_R^2(L/2)$ this test	printed Table II
1e-17	10	0.003806	0.0038
1e-17	50	0.07277	0.0728
1e-17	100	0.2593	0.259
1e-15	10	0.3806	0.381
1e-15	50	7.277	7.28
1e-15	100	25.93	25.9
1e-13	10	38.06	38.1
1e-13	50	727.7	728
1e-13	100	2593	2593

2.2 Analytic derivatives vs independent 60-digit reference (Appendix A-H)

Relative error of $d^nQ/d\eta^n$ at η_{med}, worst case over L = 20, 50, 80, 100, 120, 160 km. The reference re-implements the CAL quantities in mpmath (60 digits) and differentiates numerically; only the η-independent constants come from the engine.

quantity	n = 0	n = 1	n = 2	n = 3	n = 4
pXX	3.16e-12	1.83e-16	2.62e-16	2.10e-16	1.55e-16
eX	4.61e-11	1.03e-09	1.03e-09	1.03e-09	1.02e-09
eZ	5.25e-11	3.25e-10	3.52e-10	3.71e-10	4.23e-10
R	1.01e-10	3.17e-11	2.68e-10	2.95e-10	3.15e-10

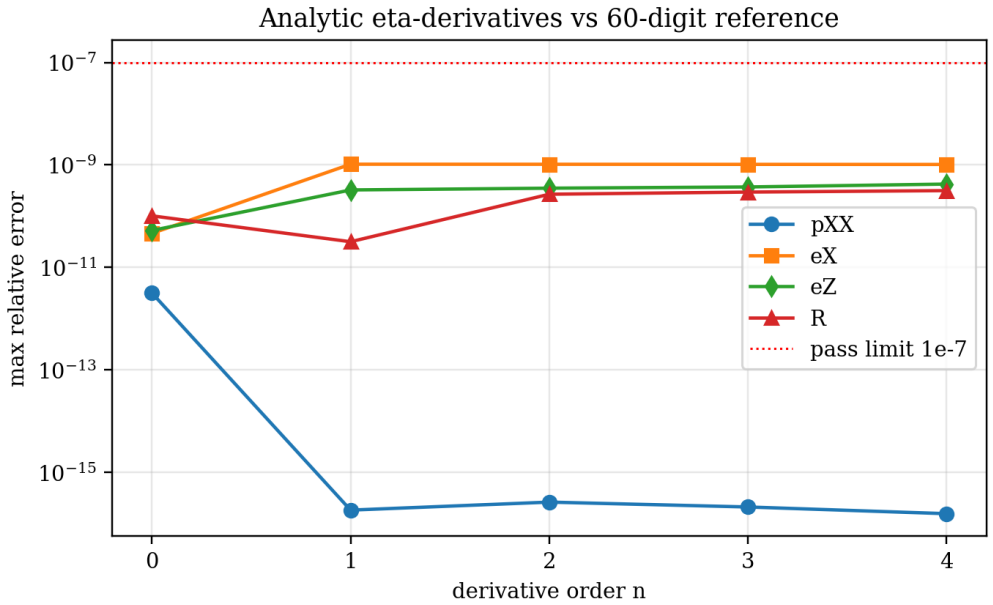


Fig. A. Derivative check. The residual ~10^{-10} is double-precision cancellation in the analytic formulas, not a formula error.

2.3 Euler operator and the two routes (Theorem 1, Sec. VI-B)

quantity	order 2: max |η-space - X-space| / |X-space|	order 4
eX	3.37e-09	3.09e-09
eZ	1.32e-09	1.90e-09
R	1.38e-11	3.11e-12

78 cross-checks (13 distances x 3 quantities x 2 orders). X-space: n-th derivative of $Q(\eta_0 e^X)$ at $X = \mu_X$ (60-digit numerical differentiation); η-space: Θ^nQ from the analytic derivatives and Stirling weights. $\Theta^n\eta^m = m^n\eta^m$ verified to 10^{-13}.

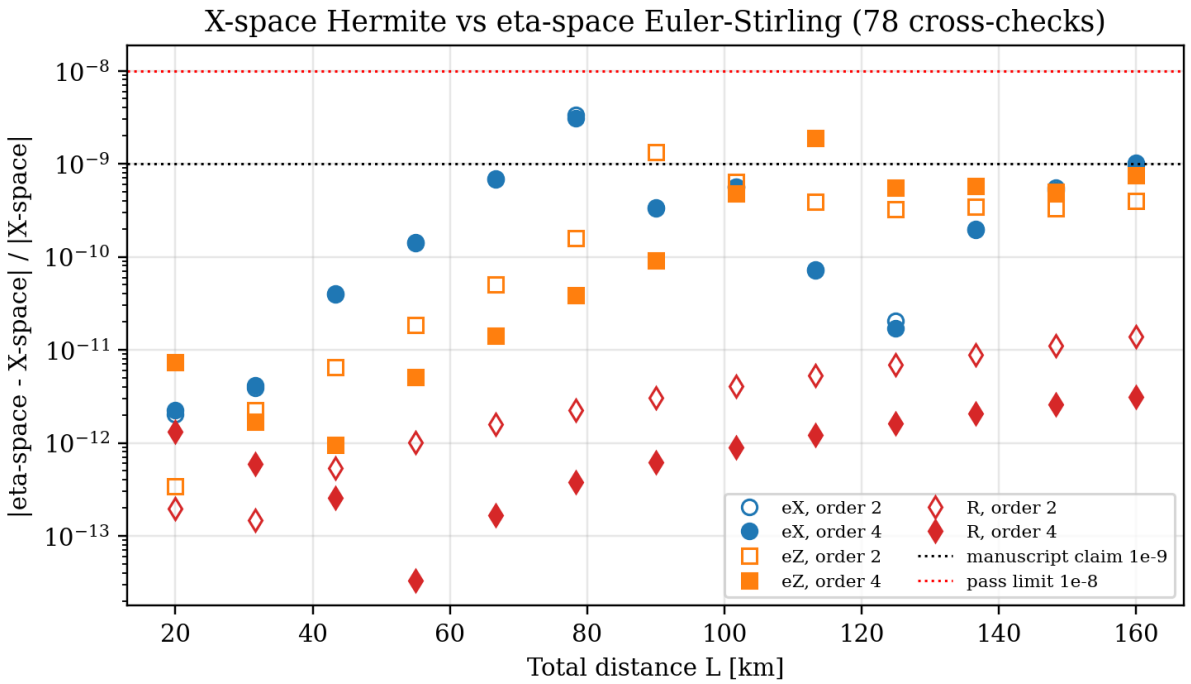


Fig. B. Agreement of the X-space Hermite and η-space Euler-Stirling formulations.

2.4 Accuracy vs Gaussian quadrature, compared with printed Tables IV, V, VI

Relative error [%] of E[Q]. "mean" = second-order delta method about E[η] in η-space; "med 2/4" = the 2-term and 4-term log-normal formulas about η_{med}. Each cell shows this test / printed / status (tolerance: 0.006 percentage points or 8 % of the printed value).

X-basis error rate e_X (Table IV)

L [km]	σ_X^2	mean: this	printed		med 2: this	printed		med 4: this	printed	
20	0.007	-0.000	+0.000	MATCH	+0.000	+0.000	MATCH	+0.000	+0.000	MATCH
50	0.036	+0.000	+0.000	MATCH	+0.000	+0.000	MATCH	+0.000	+0.000	MATCH
80	0.086	+0.000	+0.000	MATCH	-0.000	+0.000	MATCH	-0.000	+0.000	MATCH
100	0.130	+0.000	+0.000	MATCH	-0.001	-0.001	MATCH	-0.000	+0.000	MATCH
120	0.181	+0.000	+0.000	MATCH	-0.004	-0.004	MATCH	-0.000	+0.000	MATCH
160	0.307	+0.003	+0.003	MATCH	-0.053	-0.053	MATCH	-0.002	-0.003	MATCH

Z-basis phase-error rate e_Z (Table V)

L [km]	σ_X^2	mean: this	printed		med 2: this	printed		med 4: this	printed	
20	0.007	+0.000	+0.000	MATCH	+0.000	+0.000	MATCH	+0.000	+0.000	MATCH
50	0.036	+0.000	+0.000	MATCH	+0.002	+0.002	MATCH	+0.000	+0.000	MATCH
80	0.086	+0.003	+0.003	MATCH	+0.002	+0.002	MATCH	+0.000	+0.000	MATCH
100	0.130	+0.012	+0.012	MATCH	+0.001	+0.001	MATCH	+0.000	+0.000	MATCH
120	0.181	+0.034	+0.034	MATCH	-0.002	-0.002	MATCH	+0.000	+0.000	MATCH
160	0.307	+0.202	+0.202	MATCH	-0.023	-0.023	MATCH	-0.000	+0.000	MATCH

secret key rate R (Table VI)

L [km]	σ_X^2	mean: this	printed		med 2: this	printed		med 4: this	printed	
20	0.007	-0.001	-0.001	MATCH	-0.005	-0.005	MATCH	-0.000	+0.000	MATCH
50	0.036	-0.001	-0.001	MATCH	-0.044	-0.044	MATCH	-0.001	-0.001	MATCH
80	0.086	+0.002	+0.002	MATCH	-0.130	-0.130	MATCH	-0.004	-0.004	MATCH
100	0.130	+0.008	+0.008	MATCH	-0.248	-0.248	MATCH	-0.008	-0.008	MATCH
120	0.181	+0.025	+0.025	MATCH	-0.453	-0.453	MATCH	-0.016	-0.016	MATCH
160	0.307	+0.185	+0.184	MATCH	-1.341	-1.341	MATCH	-0.072	-0.072	MATCH

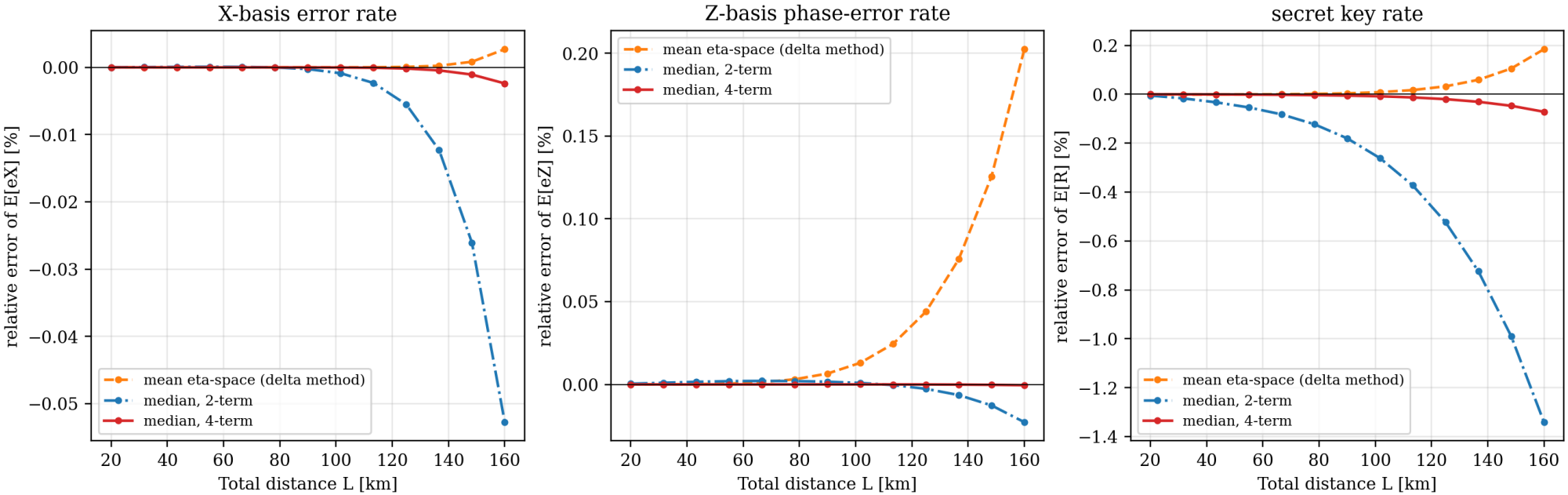


Fig. C. Error of the three averaging formulas vs distance on the 13-point sweep (compare manuscript Fig. 3). The 4-term formula stays below 0.09 % for R.

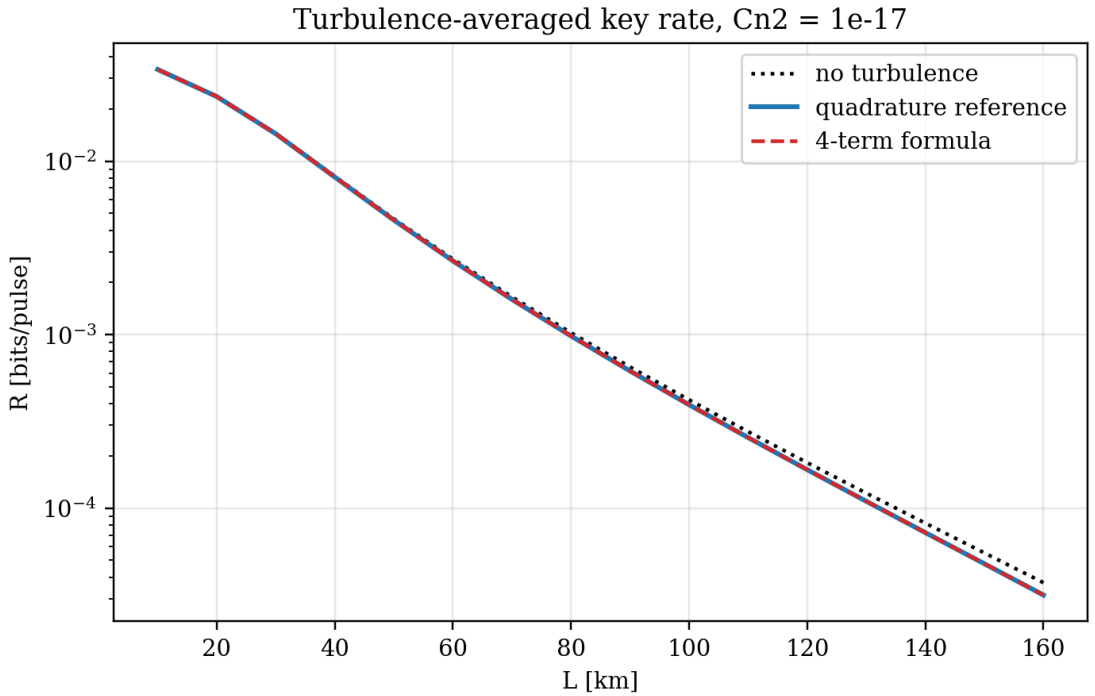


Fig. D. Turbulence-averaged key rate: quadrature reference and 4-term formula are indistinguishable on this scale.

2.5 Moments, skewness and the Appendix-H claims

case	σ_X^2 (this test)	skewness (this test)	skewness (Appendix H)
L=60, Cn2=1e-17	0.051	0.697	0.7
L=160, Cn2=1e-17	0.307	2.01	2.0
L=140, Cn2=1e-16	2.403	41.4	41.0

case	σ_X^2	naive-method error % (this test)	Appendix H %
sigma_X2=0.007 (L=20)	0.007	-0.001	-0.001
sigma_X2=0.31 (L=160)	0.307	+0.185	+0.18
Cn2=1e-16,L=140	2.403	+28.836	+29.0
total-eta,L=160	0.307	-20.133	-20.1

$E[\eta^n] = \eta_0^n \exp(n(n-2)\sigma_X^2/2)$ (Eq. 19) and the skewness (Eq. 25) agree with numerical moments to better than 10^{-6}. The naive expansion in the total transmittance $\eta_A\eta_B$ gives -20.1 % at 160 km, as printed.

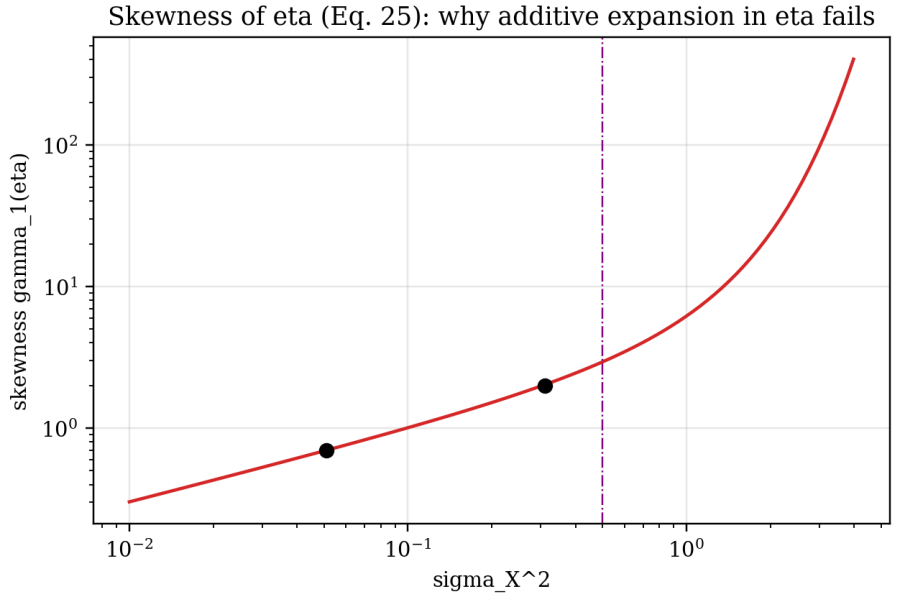


Fig. E. Skewness of η (Eq. 25). Dots: Appendix H values at σ_X^2 = 0.051 and 0.31.

2.6 Stress test beyond the weak regime (L = 80 km)

σ_X^2	arm σ_R^2	R quadrature	delta [%]	2-term [%]	4-term [%]
0.01	0.02	1.019e-03	+0.00	-0.00	-0.000
0.02	0.04	1.014e-03	+0.00	-0.01	-0.000
0.05	0.1	9.993e-04	+0.00	-0.04	-0.001
0.1	0.2	9.751e-04	+0.00	-0.17	-0.006

σ_X^2	arm σ_R^2	R quadrature	delta [%]	2-term [%]	4-term [%]
0.2	0.4	9.284e-04	+0.01	-0.66	-0.041
0.307	0.614	8.811e-04	+0.03	-1.48	-0.136
0.5	1	8.018e-04	+0.08	-3.61	-0.513
0.75	1.5	7.101e-04	+0.20	-7.30	-1.464
1	2	6.293e-04	+0.41	-11.72	-2.963
1.5	3	4.952e-04	+1.32	-21.79	-7.448
2	4	nan	+nan	+nan	+nan
3	6	nan	+nan	+nan	+nan

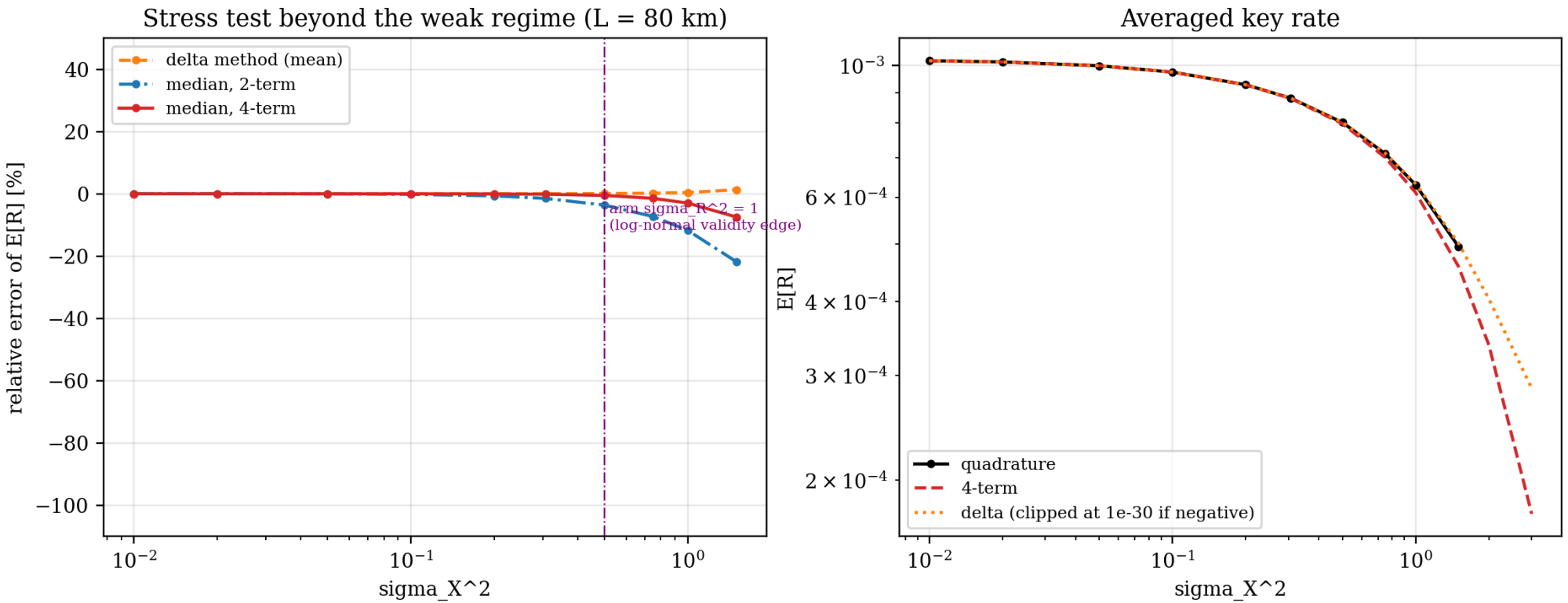


Fig. F. Beyond arm $\sigma_R^2 = 1$ (dash-dot) the log-normal model itself stops being valid, so this is a stress test of the algebra, not a physical prediction.

2.7 Monte Carlo check of the reference

L [km]	quantity	quadrature	Monte Carlo	std. error	(MC-quad)/quad	|z|
20	eX	1.95437e-02	1.95438e-02	6.0e-08	+0.0006 %	1.86
20	eZ	7.11234e-02	7.11355e-02	6.5e-06	+0.0170 %	1.86
20	R	2.35498e-02	2.35420e-02	4.2e-06	-0.0331 %	1.86
80	eX	2.00064e-02	2.00064e-02	2.9e-08	-0.0001 %	0.98
80	eZ	1.21412e-01	1.21409e-01	2.6e-06	-0.0022 %	1.05
80	R	9.81756e-04	9.82321e-04	4.9e-07	+0.0576 %	1.15
160	eX	2.10819e-02	2.10820e-02	1.0e-06	+0.0005 %	0.11
160	eZ	1.47250e-01	1.47250e-01	1.3e-05	-0.0002 %	0.02
160	R	3.15445e-05	3.15515e-05	3.4e-08	+0.0221 %	0.20

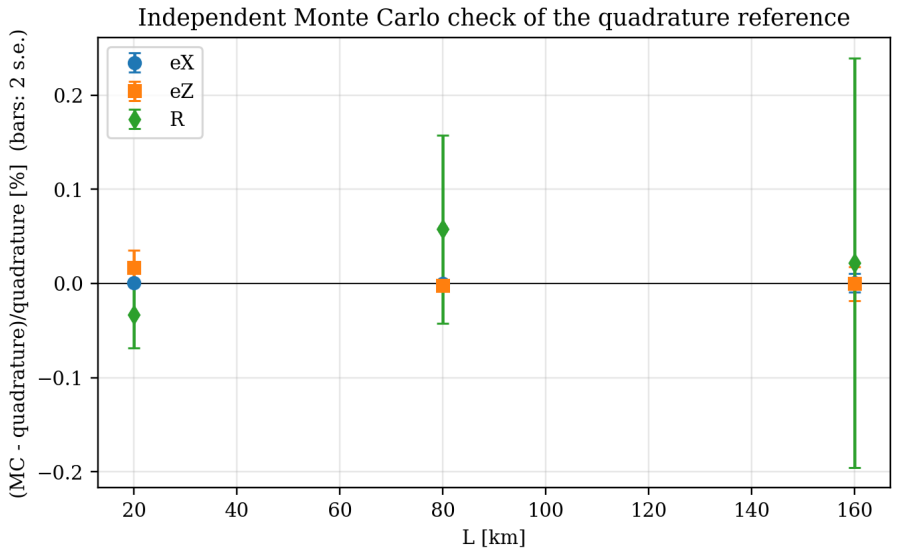


Fig. G. Independent Monte Carlo vs quadrature.

3. Gamma-Gamma extension (progress report)

Formula under test: $E[Q] = Q(\eta_{med}) + (\kappa_2/2)\Theta^2 Q + (\kappa_3/6)\Theta^3 Q + ((\kappa_4+3\kappa_2^2)/24)\Theta^4 Q$, with the log-Gamma cumulants $\kappa_1 = \psi(k) - \ln k$, $\kappa_n = \psi^{(n-1)}(k)$ summed over the two factors, aperture-averaged $\beta = 1/\sigma_y^2(D)$, single cumulative channel $\eta_{tot} = \eta_{FSO}(L)\cdot I$, Rytov variance of the full link. Two parameter conventions are run side by side: **correct** ($\sigma_R^{12/5}$ = (variance)$^{6/5}$) and **report** (variance)$^{12/5}$, which is what the report's figures and tables use (finding 1).

3.1 Cumulants and reduction to log-normal

k	κ_1 rel. err	κ_2	κ_3	κ_4
2.0	1.027e-15	0.000e+00	4.670e-15	2.428e-14
5.3	4.279e-16	5.348e-16	1.454e-15	8.034e-15
11.4	1.247e-15	2.573e-15	2.478e-15	4.540e-14
30.0	1.739e-14	6.346e-15	6.984e-15	2.896e-13

Polygamma cumulants vs direct numerical integration of ln-Gamma moments. Setting $\kappa_3 = \kappa_4 = 0$ reproduces the log-normal formula to machine precision (check G1).

3.2 Scintillation index and Gamma-Gamma parameters (Figs 1-2)

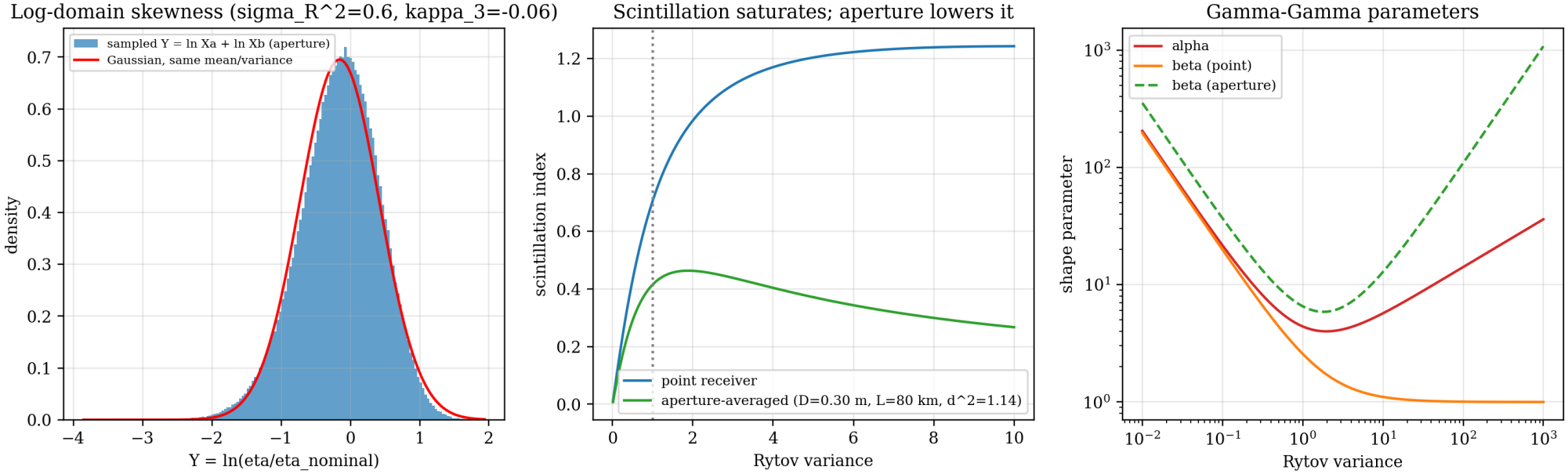


Fig. H. Left: log-domain skewness at Rytov variance 0.6 ($\kappa_3 < 0$). Middle: scintillation index, correct convention (peak ~1.2-1.3 then saturation toward 1). Right: α, β. The report's Fig. 2 corresponds to the variance$^{12/5}$ convention (see Section 1, finding 1).

3.3 Table 1: accuracy of the Gamma-Gamma formulas, aperture-averaged (D = 0.30 m)

Relative error [%] of E[R] vs the Laguerre quadrature reference. For every column: correct exponent / report convention / printed; status compares the report convention with the printed value (tolerance 0.25 pp). Rows 80* hold η_0 and d^2 at 80 km and set the Rytov variance directly.

L [km]	σ_R^2	α / β (correct)	α / β (report)	aperture gain: correct	report	printed		delta method: correct	report	printed	
20	0.05	43.17/161.63	41.76/157.41	+0.129	+0.130	+0.68	DIFFERS	-0.002	-0.002	+0.36	DIFFERS
50	0.26	9.43/19.73	7.78/17.12	+0.844	+0.890	+0.72	MATCH	-0.080	-0.112	-0.09	MATCH
80	0.61	5.34/8.26	4.21/6.88	+1.828	+1.964	+1.77	MATCH	-0.370	-0.572	-0.52	MATCH
100	0.92	4.50/5.97	4.24/5.68	+2.532	+2.585	+2.40	MATCH	-0.627	-0.703	-0.65	MATCH
120	1.29	4.14/4.84	5.32/6.09	+3.246	+2.936	+2.77	MATCH	-0.865	-0.527	-0.49	MATCH
160	2.19	4.00/3.88	10.47/10.89	+4.696	+2.905	+2.70	MATCH	-1.197	-0.155	-0.14	MATCH
80*	3.00	4.12/6.34	17.42/36.42	+6.652	+2.607	+2.36	MATCH	-0.634	-0.025	-0.02	MATCH
80*	5.00	4.58/7.95	42.19/144.55	+8.457	+1.710	+1.55	MATCH	-0.454	-0.003	-0.00	MATCH

L [km]	σ_R^2	α / β (correct)	α / β (report)	2nd order: correct	report	printed		3rd order: correct	report	printed		4th order: correct	report	printed	
20	0.05	43.17/161.63	41.76/157.41	-0.001	-0.001	+0.36	DIFFERS	-0.006	-0.007	+0.35	DIFFERS	+0.000	+0.000	+0.36	DIFFERS
50	0.26	9.43/19.73	7.78/17.12	-0.001	+0.000	-0.03	MATCH	-0.055	-0.079	-0.15	MATCH	+0.002	+0.003	+0.01	MATCH
80	0.61	5.34/8.26	4.21/6.88	+0.021	+0.036	-0.02	MATCH	-0.121	-0.192	-0.34	MATCH	+0.006	+0.012	+0.02	MATCH
100	0.92	4.50/5.97	4.24/5.68	+0.040	+0.046	-0.00	MATCH	-0.171	-0.193	-0.31	MATCH	+0.010	+0.012	+0.02	MATCH
120	1.29	4.14/4.84	5.32/6.09	+0.057	+0.035	+0.01	MATCH	-0.221	-0.132	-0.18	MATCH	+0.014	+0.007	+0.01	MATCH
160	2.19	4.00/3.88	10.47/10.89	+0.084	+0.011	+0.01	MATCH	-0.311	-0.037	-0.04	MATCH	+0.023	+0.001	+0.00	MATCH
80*	3.00	4.12/6.34	17.42/36.42	+0.037	+0.002	-0.00	MATCH	-0.211	-0.009	-0.02	MATCH	+0.014	+0.000	+0.00	MATCH
80*	5.00	4.58/7.95	42.19/144.55	+0.032	+0.001	+0.00	MATCH	-0.154	-0.001	-0.00	MATCH	+0.009	+0.000	+0.00	MATCH

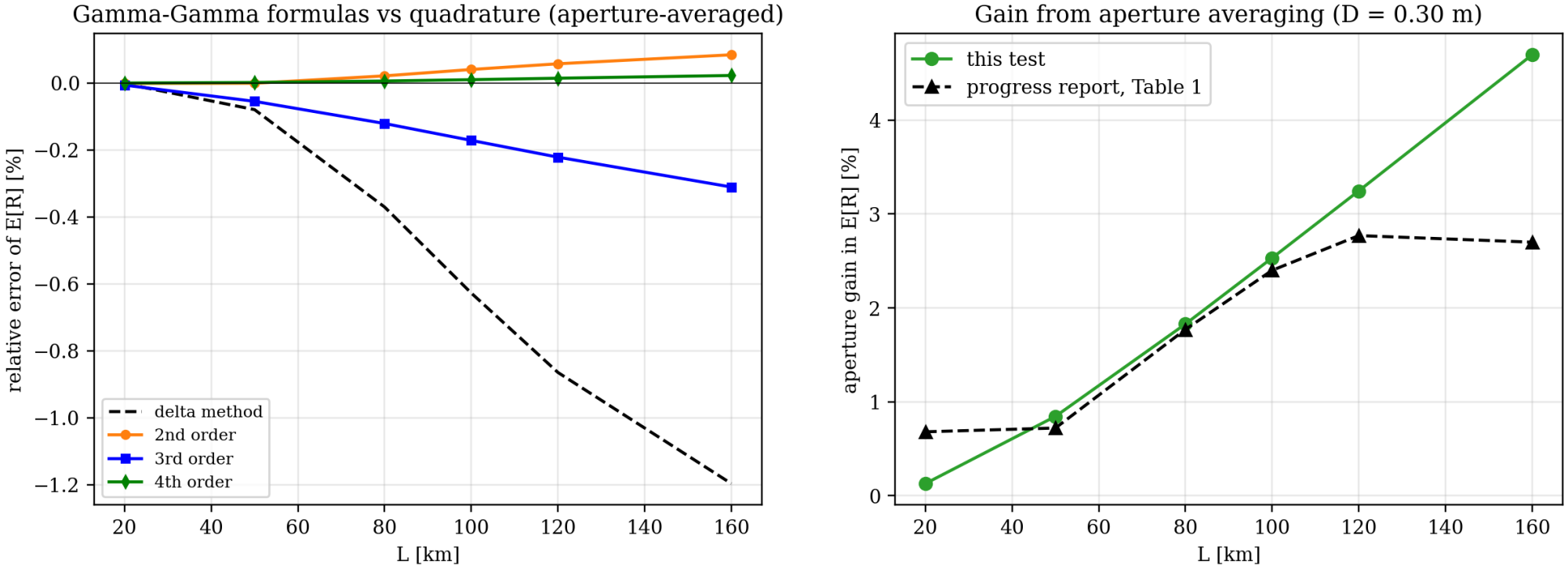


Fig. I. Left: errors of the delta method and of the 2nd/3rd/4th-order formulas vs distance (correct exponent). Right: aperture gain, this test vs the report.

3.4 Table 2: cost of using the log-normal formula on a Gamma-Gamma channel

L [km]	σ_R^2	GG 4th: correct	report	printed		LN 2nd: correct	report	printed		LN 4th: correct	report	printed	
20	0.05	+0.000	+0.000	+0.36	DIFFERS	-0.148	-0.142	+0.27	MATCH	-0.132	-0.125	+0.30	MATCH
50	0.26	+0.002	+0.003	+0.01	MATCH	-1.271	-0.939	-0.92	MATCH	-1.141	-0.808	-0.66	MATCH
80	0.61	+0.006	+0.012	+0.02	MATCH	-4.067	-3.190	-3.23	MATCH	-3.702	-2.822	-2.59	MATCH
100	0.92	+0.010	+0.012	+0.02	MATCH	-7.207	-6.948	-6.92	MATCH	-6.538	-6.278	-5.91	MATCH
120	1.29	+0.014	+0.007	+0.01	MATCH	-11.451	-12.523	-12.31	MATCH	-10.300	-11.385	-10.83	MATCH
160	2.19	+0.023	+0.001	+0.00	MATCH	-23.376	-26.558	-25.02	MATCH	-20.511	-23.812	-22.33	MATCH
80*	3.00	+0.014	+0.000	+0.00	MATCH	-32.495	-35.052	-35.60	MATCH	-28.249	-30.966	-29.83	MATCH
80*	5.00	+0.009	+0.000	+0.00	MATCH	-52.611	-54.396	-55.44	MATCH	-45.993	-48.028	-47.63	MATCH

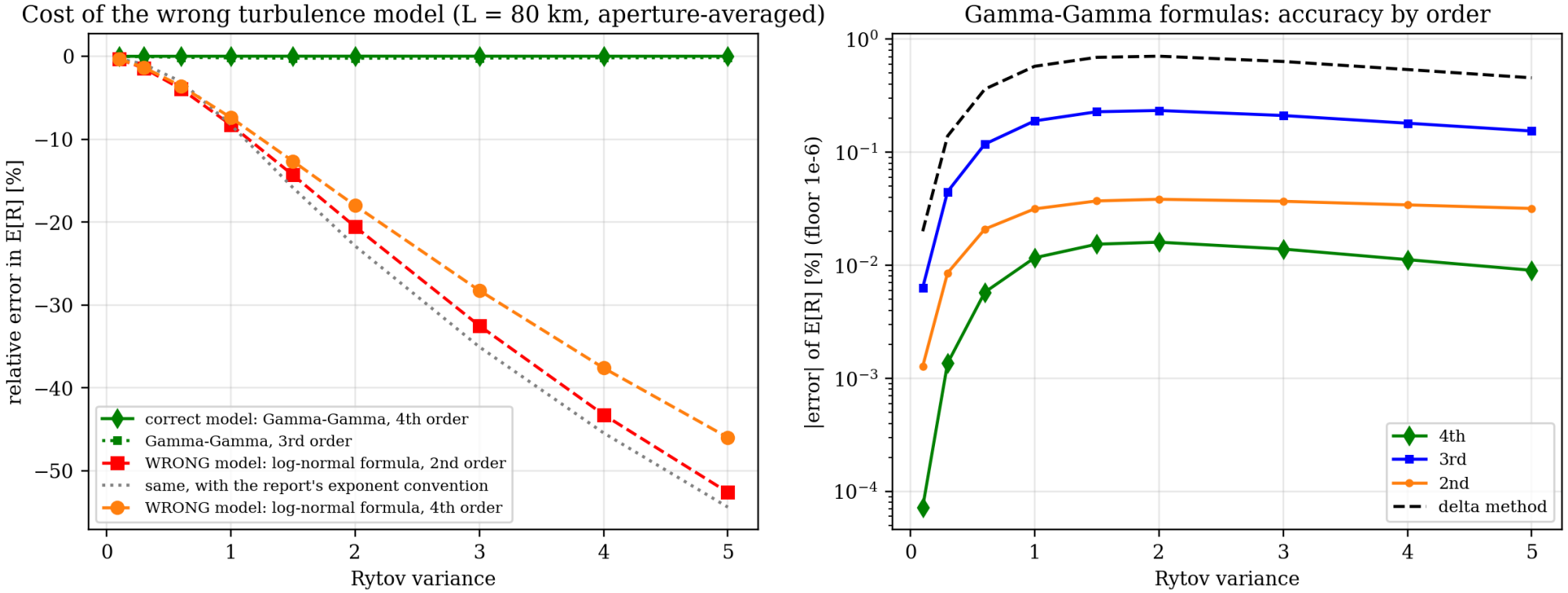


Fig. J. Compare report Fig. 3. Green: Gamma-Gamma formulas stay near zero for all Rytov variances; red/orange: the log-normal formula applied to a Gamma-Gamma channel; gray dots: same with the report's exponent convention.

3.5 Choice of expansion point (Sec. 4.3)

L [km]	σ_R^2	log-mean 2nd	log-mean 4th	mean 2nd	mean 4th	mode 2nd	mode 4th
20	0.05	-0.0013	+0.0000	+0.0097	+0.0002	-0.0159	-0.0001
50	0.26	-0.0006	+0.0017	+0.1447	+0.0060	-0.2098	-0.0040
80	0.61	+0.0214	+0.0060	+0.4350	+0.0213	-0.6146	-0.0147
100	0.92	+0.0404	+0.0100	+0.6757	+0.0351	-0.9711	-0.0250
120	1.29	+0.0575	+0.0144	+0.9093	+0.0503	-1.3319	-0.0369
160	2.19	+0.0843	+0.0225	+1.3026	+0.0785	-1.9453	-0.0592
80*	3.00	+0.0368	+0.0139	+0.7607	+0.0507	-1.0956	-0.0351
80*	5.00	+0.0318	+0.0090	+0.5563	+0.0313	-0.7730	-0.0206

Error [%] of E[R] (correct exponent) when the expansion is centred at exp(E[ln η]), at E[η], or at the mode of the Gamma-Gamma density.

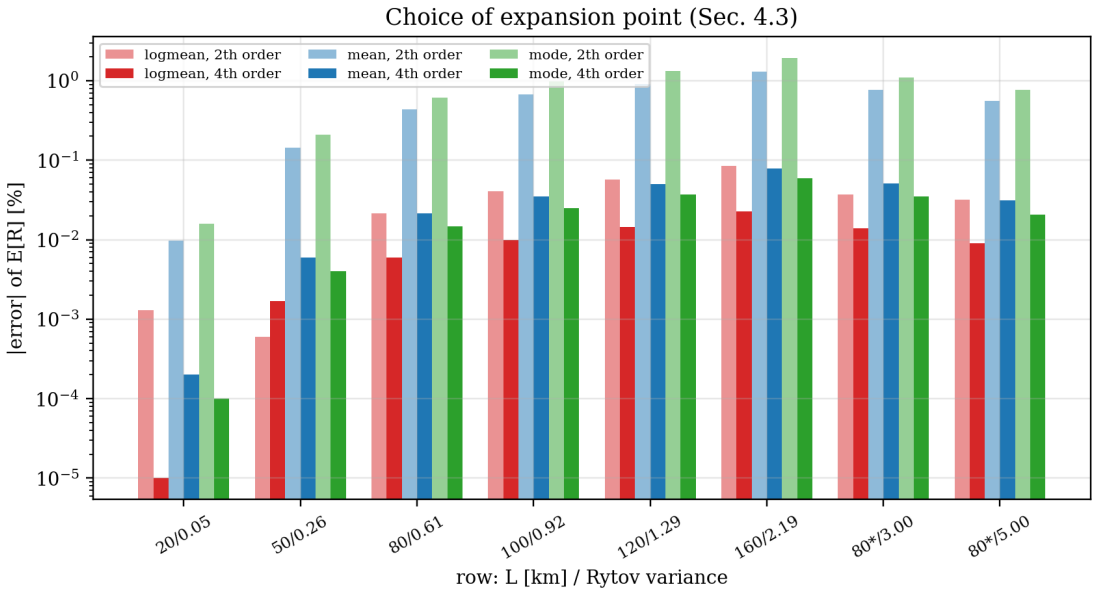


Fig. K. The log-mean point is best in every row.

3.6 Quality of the three references

L [km]	σ_R^2	α	β	R Laguerre	R Bessel-K	Bessel vs Lag. [%]	pdf norm.	R Monte Carlo	MC vs Lag. [%]	|z|
20	0.05	43.2	161.6	2.35460e-02	2.35460e-02	4.6e-12	1.000000000	2.35432e-02	-0.012	0.57
50	0.26	9.4	19.7	4.56473e-03	4.56473e-03	4.6e-13	1.000000000	4.56371e-03	-0.022	0.55
80	0.61	5.3	8.3	9.86184e-04	9.86184e-04	-1.0e-10	1.000000000	9.85895e-04	-0.029	0.54
100	0.92	4.5	6.0	4.00504e-04	4.00504e-04	-1.9e-09	1.000000000	4.00264e-04	-0.060	0.98
120	1.29	4.1	4.8	1.72485e-04	1.72485e-04	-7.0e-09	1.000000000	1.72388e-04	-0.056	0.85
160	2.19	4.0	3.9	3.47546e-05	3.47546e-05	-1.1e-08	1.000000000	3.47219e-05	-0.094	1.24
80*	3.00	4.1	6.3	9.75159e-04	9.75159e-04	-7.9e-09	1.000000000	9.74784e-04	-0.038	0.62
80*	5.00	4.6	7.9	9.81881e-04	9.81881e-04	-1.5e-09	1.000000000	9.81413e-04	-0.048	0.83

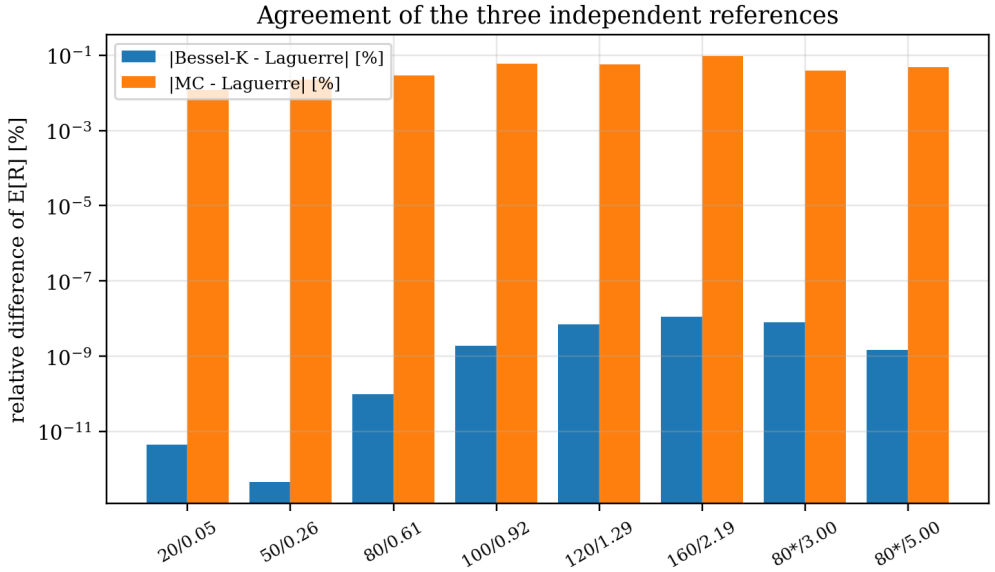


Fig. L. The tensor Laguerre rule, the Bessel-K density quadrature and Monte Carlo agree, so the references are not the source of the L=20 km offset in the printed tables.

4. Complete list of checks

PASS/FAIL checks have a numeric limit (value must be <= limit, or >= limit when marked ">="). INFO rows record findings that are not a numeric pass/fail.

01 log-normal

id	check	value	limit	status
T0-validity-1e-17	Validity distance, Cn2=1e-17, vs printed Table I *(rel. diff)*	1.54e-05	≤ 0.005	PASS
T0-eq21-1e-17	Closed form Eq.(21) vs exact solve, Cn2=1e-17 *(rel. diff)*	7.96e-08	≤ 0.005	PASS
T0-validity-1e-15	Validity distance, Cn2=1e-15, vs printed Table I *(rel. diff)*	0.00214	≤ 0.005	PASS
T0-eq21-1e-15	Closed form Eq.(21) vs exact solve, Cn2=1e-15 *(rel. diff)*	7.96e-08	≤ 0.005	PASS
T0-validity-1e-13	Validity distance, Cn2=1e-13, vs printed Table I *(rel. diff)*	0.00273	≤ 0.005	PASS
T0-eq21-1e-13	Closed form Eq.(21) vs exact solve, Cn2=1e-13 *(rel. diff)*	7.96e-08	≤ 0.005	PASS
T0-tab2-1e-17-10	Per-arm Rytov variance, Cn2=1e-17, L=10 km, vs Table II *(rel. diff)*	0.00167	≤ 0.01	PASS
T0-tab2-1e-17-50	Per-arm Rytov variance, Cn2=1e-17, L=50 km, vs Table II *(rel. diff)*	0.000415	≤ 0.01	PASS
T0-tab2-1e-17-100	Per-arm Rytov variance, Cn2=1e-17, L=100 km, vs Table II *(rel. diff)*	0.00124	≤ 0.01	PASS
T0-tab2-1e-15-10	Per-arm Rytov variance, Cn2=1e-15, L=10 km, vs Table II *(rel. diff)*	0.000964	≤ 0.01	PASS
T0-tab2-1e-15-50	Per-arm Rytov variance, Cn2=1e-15, L=50 km, vs Table II *(rel. diff)*	0.000415	≤ 0.01	PASS
T0-tab2-1e-15-100	Per-arm Rytov variance, Cn2=1e-15, L=100 km, vs Table II *(rel. diff)*	0.00124	≤ 0.01	PASS
T0-tab2-1e-13-10	Per-arm Rytov variance, Cn2=1e-13, L=10 km, vs Table II *(rel. diff)*	0.000964	≤ 0.01	PASS
T0-tab2-1e-13-50	Per-arm Rytov variance, Cn2=1e-13, L=50 km, vs Table II *(rel. diff)*	0.000415	≤ 0.01	PASS
T0-tab2-1e-13-100	Per-arm Rytov variance, Cn2=1e-13, L=100 km, vs Table II *(rel. diff)*	8.55e-05	≤ 0.01	PASS
T1-deriv-pXX	Analytic derivatives of pXX, orders 0-4, vs 60-digit reference (max over 6 distances) *(max rel. error)*	3.16e-12	≤ 1e-07	PASS
T1-deriv-eX	Analytic derivatives of eX, orders 0-4, vs 60-digit reference (max over 6 distances) *(max rel. error)*	1.03e-09	≤ 1e-07	PASS
T1-deriv-eZ	Analytic derivatives of eZ, orders 0-4, vs 60-digit reference (max over 6 distances) *(max rel. error)*	4.23e-10	≤ 1e-07	PASS
T1-deriv-R	Analytic derivatives of R, orders 0-4, vs 60-digit reference (max over 6 distances) *(max rel. error)*	3.15e-10	≤ 1e-07	PASS
T1-engine-vs-reference	High-precision reference reproduces the validated engine (value, L=80 km) *(max rel. error)*	1.11e-12	≤ 1e-09	PASS
T1-pxx-erratum	**Printed Eq.(6)/Appendix A bracket for pXX**: Eq.(6) as printed gives pXX = 0.5 at L=80 km, whereas CAL Eq.(30) / the engine give 0.0015695. The A^2 term must sit outside the factor 1/2: pXX = (A/2)[e^{-(1-O)g}+e^{-(1+O)g}] - A^2 e^{-2g}. With this form all derivatives agree with the reference (check T1-deriv-pXX); the same bracket slip appears in the printed pXX derivatives.			INFO
T1-Dj-note	**Truncation residual D_j**: D_j (CAL Delta_j) is eta-independent in CAL and in the engine (residual_j), so D_j^(n)=0 for n>=1; Appendix E says D_j^(n) "follows identically" from the sqrt(pZZ) terms, which would only hold for a different D_j. The analytic derivatives with D_j constant match the reference to ~1e-10.			INFO
T2-euler	Theta^n eta^m = m^n eta^m via Stirling weights (n=1..4, m=1..6) *(max rel. error)*	2e-16	≤ 1e-13	PASS
T3-two-routes	X-space vs eta-space formulation (78 cross-checks); manuscript claims < 1e-9 *(max rel. diff)*	3.37e-09	≤ 1e-08	PASS
T3-count	**Number of cross-checks**: 78 (13 distances x 3 quantities x 2 orders)			INFO
T4-sigmaX2-20	sigma_X^2 at L=20 km vs printed *(abs. diff)*	0.000218	≤ 0.0006	PASS
T4-sigmaX2-50	sigma_X^2 at L=50 km vs printed *(abs. diff)*	0.000385	≤ 0.0006	PASS
T4-sigmaX2-80	sigma_X^2 at L=80 km vs printed *(abs. diff)*	0.000127	≤ 0.0006	PASS
T4-sigmaX2-100	sigma_X^2 at L=100 km vs printed *(abs. diff)*	0.000339	≤ 0.0006	PASS
T4-sigmaX2-120	sigma_X^2 at L=120 km vs printed *(abs. diff)*	0.000124	≤ 0.0006	PASS
T4-sigmaX2-160	sigma_X^2 at L=160 km vs printed *(abs. diff)*	7.69e-05	≤ 0.0006	PASS

id	check	value	limit	status
T4-reproduce-tables	Entries of printed Tables IV-VI reproduced (54 of 54) *(fraction)*	1	≥ 0.95	PASS
T4-claim-4th-order-R	Claim: 4-term R error < 0.09 % over the sweep (checked on 13-point sweep below) *(max |error| %, Table rows)*	0.0716	≤ 0.09	PASS
T4-reference-GH-vs-quad	Reference: Gauss-Hermite (200 nodes) vs adaptive quad (rel tol 1e-11), R at L=160 km *(rel. diff)*	3.21e-11	≤ 1e-09	PASS
T4-claim-13pt-R4	Claim: 4-term secret-key-rate error < 0.09 % over the full 13-point sweep *(max |error| %)*	0.0716	≤ 0.09	PASS
T5-moments	E[eta^n] = eta0^n exp(n(n-2) s^2/2) (Eq. 19) vs numerical, n=1..4 *(max rel. error)*	1.2e-15	≤ 1e-09	PASS
T5-skewness	Skewness of eta, Eq.(25), vs numerical *(max rel. error)*	2.38e-14	≤ 1e-06	PASS
T5-skew-60	Appendix H: gamma_1 ~ 0.70 at L=60 km *(abs. diff)*	0.003	≤ 0.02	PASS
T5-skew-160	Appendix H: gamma_1 ~ 2.0 at L=160 km *(abs. diff)*	0.01	≤ 0.1	PASS
T5-skew-140	**Appendix H: "gamma_1 ~ 41 at sigma_X^2 ~ 2.4 (Cn2=1e-16, L=140)"**: for Cn2=1e-16, L=140 km this test gives sigma_X^2 = 2.40 and gamma_1 = 41.4, matching the printed 41. The "sigma_X^2 = 9.6" quoted for the same case in the next sentence equals sigma_A^2+sigma_B^2 = 4 sigma_X^2, so it is a typo for 2.4.			INFO
T5-naive-0.007	Appendix H: naive error -0.001 % at sigma_X^2=0.007 *(abs. diff, pp)*	2.84e-05	≤ 0.003	PASS
T5-naive-0.31	Appendix H: naive error 0.18 % at sigma_X^2=0.31 *(abs. diff, pp)*	0.00451	≤ 0.01	PASS
T5-naive-140	**Appendix H: naive error "29 %" at Cn2=1e-16, L=140**: this test: +28.8 % at sigma_X^2 = 2.40: the 29 % is reproduced, but the printed sigma_X^2 = 9.6 should read 2.4.			INFO
T5-naive-total	**Appendix H: naive expansion in eta_A*eta_B errs -20.1 % at L=160**: this test: -20.1 %			INFO
T6-validity-4term	4-term R error < 0.1 % while per-arm Rytov variance <= 0.2 (sigma_X^2 <= 0.1) *(max |error| %)*	0.00552	≤ 0.1	PASS
T6-naive-negative	**Naive delta method gives a negative key rate**: never up to sigma_X^2 = 3 in the manuscript (two-arm) convention; the progress report's "negative key rate near Rytov variance two" refers to a single cumulative log-normal channel and is tested in script 02			INFO
T7-mc	Monte Carlo (400,000 samples) vs quadrature reference: max |z| *(max |z| (std. errors))*	1.86	≤ 4	PASS

02 Gamma-Gamma

id	check	value	limit	status
G0-cumulants	Polygamma cumulants of ln Gamma(k,k) vs direct numerical integration (k = 2..30, orders 1-4) *(max rel. error)*	2.9e-13	≤ 1e-05	PASS
G1-reduction	Cumulant formula with kappa_3 = kappa_4 = 0 equals the log-normal formula (2nd and 4th order) *(max rel. diff)*	0	≤ 1e-12	PASS
G2-d2	Aperture parameter d^2 = k D^2 / 4L at L=80 km, D=0.30 m (printed 1.14) *(abs. diff)*	9.41e-05	≤ 0.01	PASS
G2-fig2-point-report-convention	Report Fig. 2: point-receiver SI peak 0.71 (reproduced with variance^(12/5) convention) *(abs. diff)*	0.00193	≤ 0.01	PASS
G2-fig2-aperture-report-convention	Report Fig. 2: aperture-averaged SI peak 0.44 (same convention) *(abs. diff)*	0.000401	≤ 0.01	PASS
G2-fig2-peak-position	Report Fig. 2: SI peak at Rytov variance ~1 (same convention) *(abs. diff)*	0.0636	≤ 0.25	PASS
G2-fig2-decay-report	Report convention: SI at Rytov variance 10 falls to ~0.08 (point receiver) *(abs. diff)*	0.00104	≤ 0.02	PASS
G2-correct-saturation	Correct exponent sigma_R^(12/5): SI(10) lies in [1.1, 1.35] (saturation toward 1) *(bool)*	1	≥ 1	PASS
G2-correct-vs-report-at-10	SI at Rytov variance 10: correct / report convention (point receiver) *(ratio)*	15.7	≥ 5	PASS
G2-aperture-lowers	Aperture lowers SI at every turbulence strength (correct convention) *(bool)*	1	≥ 1	PASS
G2-exponent-finding	**Gamma-Gamma exponent in the progress report**: The report's Fig. 2 (peak 0.71 then decay to ~0.08 at Rytov variance 10) is reproduced only when sigma_R^(12/5) is evaluated as (Rytov variance)^(12/5). In the Andrews-Phillips formulas sigma_R is the Rytov STANDARD DEVIATION, i.e. the exponent is 6/5 on the variance; with it the scintillation index peaks at ~1.2-1.3 near variance 5-10 and saturates toward 1 (aperture-averaged curve scales down accordingly). The same slip was present in the repository's Gamma-Gamma code until v2.2.0. The cumulant formulas themselves are unaffected and are tested below under BOTH conventions.			INFO
G2-SI-vs-MC	SI = 1/a + 1/b + 1/(ab) vs Monte Carlo variance of I (sigma_R^2 = 0.6) *(rel. diff)*	0.00269	≤ 0.01	PASS
G2-kappa3	**Fig. 1 label "kappa_3 = -0.19" at sigma_R^2 = 0.6**: kappa_3 = -0.191 (point) / -0.094 (aperture) with the report convention; -0.132 / -0.057 with the correct exponent. Either way kappa_3 < 0: left-skewed log-domain distribution.			INFO

id	check	value	limit	status
G3-4th-order-correct	Claim: 4th-order GG formula within ~0.02 % of quadrature in all Table-1 rows (correct exponent) *(max |error| %)*	0.0225	≤ 0.03	PASS
G3-ordering-correct	Error shrinks with order, max|4th| < max|3rd| < max|delta| (correct) *(bool)*	1	≥ 1	PASS
G3-naive-modest-correct	Claim: delta method degrades gracefully for Gamma-Gamma, |error| < 2 % in all rows (correct) *(max |error| %)*	1.2	≤ 2	PASS
G3-aperture-positive-correct	Aperture averaging raises the averaged key rate in every row (correct) *(bool)*	1	≥ 1	PASS
G3-4th-order-report-conv	Claim: 4th-order GG formula within ~0.02 % of quadrature in all Table-1 rows (report-conv exponent) *(max |error| %)*	0.0124	≤ 0.03	PASS
G3-ordering-report-conv	Error shrinks with order, max|4th| < max|3rd| < max|delta| (report-conv) *(bool)*	1	≥ 1	PASS
G3-naive-modest-report-conv	Claim: delta method degrades gracefully for Gamma-Gamma, |error| < 2 % in all rows (report-conv) *(max |error| %)*	0.703	≤ 2	PASS
G3-aperture-positive-report-conv	Aperture averaging raises the averaged key rate in every row (report-conv) *(bool)*	1	≥ 1	PASS
G3-reproduce-table1	Printed Table 1 entries reproduced with the report's exponent convention, within 0.25 pp (35 of 40) *(fraction)*	0.875	≥ 0.7	PASS
G3-correct-exponent	**Table 1 with the correct exponent**: 26 of 40 printed entries are within 0.25 pp when the correct sigma_R^(12/5) is used; the differences are concentrated in the aperture-gain and delta-method columns at Rytov variance > 1 (the correct SI is larger there). The 4th-order formula stays within 0.03 % either way.			INFO
G4-ln-model-error-report	Report convention: log-normal formula loses > 50 % of the key rate at Rytov variance 5 (2nd order) *(|error| %)*	54.4	≥ 50	PASS
G4-ln-model-error-correct	Correct exponent: error of the log-normal formula at Rytov variance 5 (2nd order); still large *(|error| %)*	52.6	≥ 10	PASS
G4-order-no-fix	Raising the log-normal order does not repair the model error (|4th| > 0.5 |2nd| in the last row, correct exponent) *(bool)*	1	≥ 1	PASS
G4-reproduce-table2	Printed Table 2 entries reproduced with the report convention, tolerance max(1 pp, 10 %) (23 of 24) *(fraction)*	0.958	≥ 0.6	PASS
G4-L20-artifact	**Table 1/2, L = 20 km row**: the printed +0.36 % appears identically in the delta, 2nd, 3rd and 4th-order columns, so it does not depend on the formula. With the Laguerre, Bessel-K and Monte Carlo references used here the four errors are -0.002, -0.001, -0.006, +0.000 % (alpha, beta = 43, 162). A formula-independent offset of this kind points at the reference integral for very large shape parameters (or its normalisation), not at the expansion.			INFO
G5-4th-all-sigma	4th-order GG error < 0.05 % for Rytov variance 0.1-5 (L = 80 km) *(max |error| %)*	0.016	≤ 0.05	PASS
G5-4th-all-sigma-report	4th-order GG error < 0.05 % for Rytov variance 0.1-5 (report convention) *(max |error| %)*	0.012	≤ 0.05	PASS
G8-ln-naive-negative	**Report: naive delta applied to a log-normal channel gives a negative key rate near Rytov variance 2**: this test: the delta-method key rate first becomes <= 0 at Rytov variance ~2.30 (single cumulative channel, sigma_X^2 = sigma_R^2)			INFO
G6-logmean-best	Report: expanding at the median (log-mean point) is better than at the mean or the mode (sum of |errors|, all rows) *(bool)*	1	≥ 1	PASS
G6-terminology	**Expansion point name**: the report calls exp(E[ln eta]) the "median"; for Gamma-Gamma it is the geometric mean (log-mean), which differs slightly from the true median of eta. For the log-normal model the two coincide.			INFO
G7-laguerre-vs-bessel	Laguerre tensor quadrature vs Bessel-K density quadrature (max over rows) *(max rel. diff)*	1.12e-10	≤ 0.0001	PASS
G7-mc	Monte Carlo vs Laguerre quadrature: max |z| *(max |z|)*	1.24	≤ 4	PASS
G7-pdf-norm	Bessel-K density integrates to 1 (max |norm-1|) *(max abs. diff)*	7.04e-14	≤ 1e-06	PASS

5. Reproducing this report

python3 analytical_validation/run_all.py runs scripts 01, 02 and 03 (about 30 s; set QUICK=1 for a coarse run). Scripts 01 and 02 write results/tables/*.csv|json and results/figures/*.png; script 03 builds this PDF from them. No result is typed by hand, and the reference methods share no code with the formulas under test.